

Software Testing

Manual Testing - SwagLabs | Automation Testing - Automation Exercise

An end-to-end journey through manual QA practice on SwagLabs and a Selenium + TestNG automation framework built on Automation Exercise.

Ankita Rout

Emp ID:-85009812

Graduate Engineer Trainee

Content

- 1 What, Why & When of Testing
- 2 Manual Testing on SwagLabs
- 3 Test Plan, Scenarios & Cases
- 4 Defect Reporting & RTM
- 5 Automation Testing on Automation Exercise
- 6 Framework, Tools & Folder Structure
- 7 Reports, API Testing & Version Control

What, Why & When of Testing

What?

Testing is the process of executing a program with the intent of finding bugs, validating that the application meets client requirements and quality expectations.

Why?

Testing identifies defects early, ensures product quality and reliability, and reduces the risk of failures that impact end users and the business.

When?

Testing runs across the entire SDLC - starting at requirement analysis and continuing through development, deployment, and maintenance.

Manual Testing

SECTION 1

- Process of testing software without using automation tools
- Testers execute test cases manually, step by step, to find defects
- Helps understand the application from a real user's perspective
- Catches usability and visual issues automation often misses
- Foundation for writing strong, reliable automation scripts later

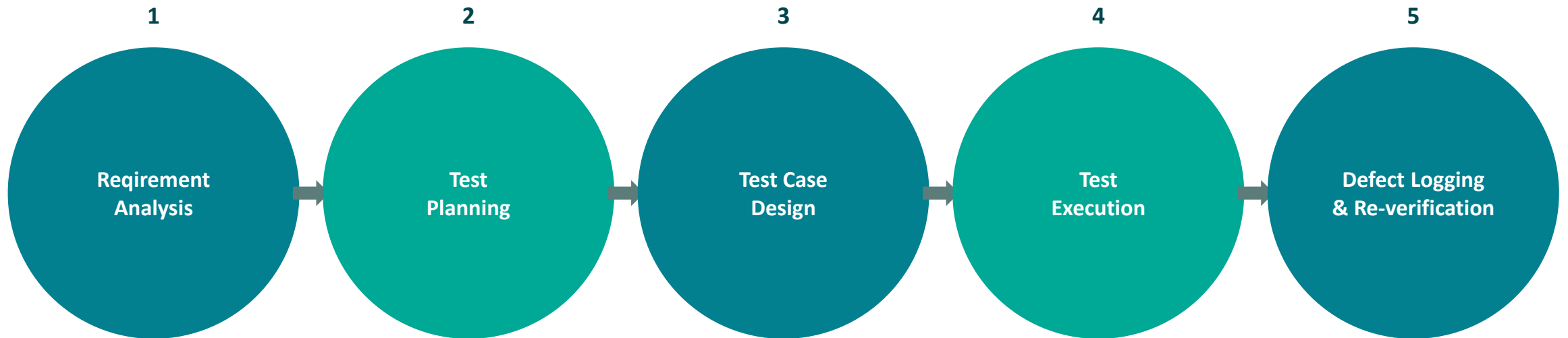
Application Under Test

SwagLabs

A demo e-commerce web application used to practice manual testing across login, product catalog, cart, and checkout flows.

- Login & authentication
- Product inventory & sorting
- Shopping cart
- Checkout & order flow

Life Cycle of Manual Testing (STLC)



Each cycle feeds back into planning - defects found during execution refine future test scenarios and cases on SwagLabs.

Test Plan

Test Plan for Swaglabs.in Website

1. Test Plan Identifier

File Name: 1.0_Swaglabs_tplan_v1.0.doc

- **Document Name:** Test Plan – E-commerce Application
- **Version:** SL_EC_1.0
- **Date:** 18-06-2026
- **Location:** Coforge Ltd, Hyderabad

2. Document History

REV	Author	Edits	Date
1.0	Ankita Rout, Graduate Engineer Trainee	Initial Draft	18/06/2026

3. Introduction

SwagLabs is a demo e-commerce web application that allows users to log in, view products, add items to the cart, and complete purchases.

This test plan defines the strategy to validate the functionality, usability, and performance of the SwagLabs application. The goal is to ensure that all core features such as login, product selection, cart management, and checkout process work correctly and provide a smooth user experience.

4. Test Items

Specifies **what components/modules** will be tested.

Example:

- User Login Module
- Product Listing Module
- Product Details Module
- Add to cart Module
- Shopping Cart Module
- Checkout Information Module
- Checkout Overview Module
- Order Completion Module
- Product Sorting Module
- Logout Functionality

5. Features to be Tested

Medium priority features are features that have been stable and High Priority features are features that have been recently changed.

Feature to be Tested	Priority	Comments
User Login	High	
Product Listing	High	
Product Details View	High	
Add to Cart	High	
Remove from Cart	High	
Shopping Cart	High	
Checkout Process	High	
Order confirmation	High	
Product Sorting	medium	
Logout	High	

6. Features NOT to be Tested

- Backend database validation.
- Third-party integrations.
- Server performance testing.
- Installation and deployment
- Experimental features

7. Test Environment

LP-AZ-710D590tt
HP ProBook 440 G7

Rename this PC

① Device specifications

Copy

Device name	LP-AZ-710D590tt
Full device name	LP-AZ-710D590tt.IN.COFOBIGETECH.COM
Processor	Intel(R) Core(TM) i7-10510U CPU @ 1.80GHz (2.30 GHz)
Installed RAM	16.0 GB (15.8 GB usable)
Device ID	85BA0706-222D-4183-856D-25FEA7DE9ACA
Product ID	00329-00000-00003-AA401
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display.

Test Plan



8. Test Approach / Test Strategy

Describes **how testing will be performed**.

The test strategy for Swaglabs1.0 will include manual testing with automated testing(**Selenium With Java**) where time/need permits. Automated testing will be used to test the basic Interface objects keyboard and mouse navigation or any redundant tasks best translated to automation. Automated testing will be limited to a Win-64 platform because of Automation tool limitations.

- **Sanity test** will include an very short battery of tests to ensure the Swaglabs1.0 runs/launches and can connect to the back end database(**MySql /SQL Server 2025**). Test execution will happen as individual component testing. These tests will be fully automated.
- **Acceptance tests** will include a short battery of tests that will only touch major features to ensure Swaglabs1.0 will operate smoothly at a basic level. These tests will be executed on **Windows 11 Enterprise**. The goal will be to automate all of these tests.
- Full Functional tests will include a **comprehensive test** of all features listed in this document. These tests will be limited to the Windows 11 enterprise. These test will include automation with time allowing.

8.1 Types of Testing

- Smoke Testing
- Sanity Testing
- Functional Testing
- Integration Testing
- System Testing
- Regression Testing

8.2 Test Levels

- Unit Testing (by developers)
- Integration Testing
- System Testing
- Acceptance Testing

8.3 Test Techniques

- Black Box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing **Functionality Testing**

Usability Testing

The usability testing will be accomplished by verifying the information in each window are accurate. Menus, icons and toolbar functionality will be tested as applicable to the navigation and results panes. Multi Window Overlapping will be tested because product supports opening of multiple documents.

Robustness/Reliability Testing

Test the GUI for correct keyboard and mouse navigation, windows, menus, buttons, etc. These tests will include Keyboard only navigation of Swaglabs1.0, Mouse only navigation. Correctness and existence of warning/error dialog boxes. The correct/expected functionality of buttons, icons and menus.

Stress Testing

Not applicable

9. Entry and Exit Criteria

9.1 Entry Criteria

Conditions to start testing:

- Requirements approved
- Test environment ready
- Test data available
- Build deployed

9.2 Exit Criteria

Conditions to stop testing:

- All planned test cases executed
- Critical and high defects closed
- Test summary report prepared

10. Test Deliverables

Test plan	1.0_Swaglabs_tplan.doc
Test Specification	1.0_Swaglabs_tspec.doc
Test Automation Plan	TBD()
Test Checklist	TBD
Test Logs	TBD
Test tools	Selenium 4.38, Postman, Rest Assured, Jmeter, Jenkins, Git and Git

Hub and Appium and MySql Server 2025

11. Testing Tasks & Resource Needs

- Use of MySql Server database.
- Write Test Specification.
- Write Test Automation plan.
- Write Test Checklist.
- Setup workspace.
- Develop Automation script.

➤ Minimum requirements for platforms:

Test Plan

12. Schedule

Test Documentation

Document	Development Time		Sign-off Date	Dependencies
	Person-Days	Completion Date		
Test Plan	1 day	18/06/2026	18/06/2026	Functional Spec. Review
Test Specifications	1 day	18/06/2026	18/06/2026	Test plan, Functional Spec, Review
Test Logs	TBD	18/06/2026		
Test Summary Report	TBD	18/06/2026		
Checklist	TBD	18/06/2026		

Test Development

Deliverable	Development Time		Dependencies
	Person-Days	Completion Date	
Functional Test Suite	1	18/06/2026	
Automation build			

Test Execution

Test Activity		Execution Time (Person-Days)			Dependencies
		Alpha	Beta	FCS	
Acceptance		1 Day	1 Day	1Days	
Functional	Functionality	1	1		
	Usability	All High to Medium test cases			
	Robustness				
	Stress				
	Multi Inst.				
Regression					
Documentation Review		1	1	1	
Total Test Cycle		1	1	1	

13. Risks and Assumptions

Risks:

- **Environment Instability:** Delays in setting up the test environment could impact the schedule- VM is not working port no.-5103
- **Resource Availability:** Limited availability of testers could delay test execution.

- **Late Requirement Changes:** Changes in requirements during the testing phase could lead to additional testing efforts.
- **Contingencies:** Backup resources and alternate test environments to handel unexpected delays or system failures.
- **Backup Resources:** Have backup resources ready to join the testing effort if required.
- **Flexible Schedule:** Allocate buffer time in the schedule to accommodate unexpected delays.
- **Regular Communication:** Hold daily stand-up meetings to quickly address any issues that arise

14. Approvals \Sign of

POSITION	NAME	SIGNATURE	REVIEWED	APPROVAL
*Approval required Project Manager				
Project Manager			Raghavendra BN	Raghavendra BN
Client Approval	Coforge			

Test Scenarios

Name: Ankita Rout Email ID: routankita293@gmail.com Emp ID: 85009812				
TEST SCENARIOS FOR SWAGLABS.COM				
Scenario ID	Scenario Name	Scenario Description	Preconditions	Expected Outcome
TS_01	Verify User Login Module	Validate login with valid and invalid credentials	User registered	User logged in or error message displayed
TS_02	Verify Product Listing Module	Validate display of all products on inventory page	User logged in	Products listed correctly
TS_03	Verify Product Details Module	Validate product details view	User on product page	Product details displayed correctly
TS_04	Verify Product Sorting Module	Validate sorting of products (price/name)	User logged in	Products sorted correctly
TS_05	Verify Add to Cart Module	Validate adding products to cart	Product available	Product added to cart successfully
TS_06	Verify Remove from Cart Module	Validate removing products from cart	Product in cart	Product removed successfully
TS_07	Verify Shopping Cart Module	Validate cart page contents and navigation	Items added to cart	Cart items displayed correctly
TS_08	Verify Checkout Information Module	Validate entering user details during checkout	Items in cart	Correct order summary displayed
TS_09	Verify Checkout Overview Module	Validate order summary before payment	Checkout step initiated	Correct order summary displayed
TS_10	Verify Order Completion Module	Validate successful order placement	Checkout completed	Order placed successfully
TS_11	Verify Logout Functionality	Validate logout from application	User logged in	User logged out and redirected to login page

Test Cases

A	B	C	D	E	F	G	H
Project Name:	E-commerce Application						
Module Name:	SwagLabs Website	Date Updated:	18-06-2026				
Last Build Tested:		Reviewed By:	Raghavendra BN				
Run No:	9 Test Cases	Executed By:	Ankita Rout				
Run Date:	18-06-2026						
Test Objectives:	Writing/Designing test cases for SwagLabs						
Test Setup:	MS Excel Sheet (Online)						
Req_Id	Test_Case ID	Test_Case_Descriptio	Test_steps	Test_Data	Expected_Result	Actual_Result	Status
Req_1.0	TC_01	Verify user can login with valid credentials	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should enter valid username 4. User should enter valid password 5. User should click on Login button	Url:https://www.saucedemo.com/ Username: standard_user Password: secret_sauce	User should be logged in successfully and redirected to inventory page	Logged in successfully	Passed
Req_1.1	TC_02	Verify products are displayed	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3. User should login to application 4. User should navigate to	Url: https://www.naukri.com/	All products should be visible	Products displayed successfully	Passed

Test Cases

Req_1.2	TC_03	Verify user can add product to cart	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3. User should login. 4. Select any product 5. Click on "Add to Cart" button 6. Navigate to cart	Product: Backpack	Product should be added to cart successfully	Product added successfully	Passed
Req_1.3	TC_04	Verify user can remove product from cart	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should login 4.Add product to cart 5.Navigate to cart page 6.Click on "Remove" button	Product: Bike Light	Product should be removed from cart	Product removed successfully	Passed
Req_1.4	TC_05	Verify product sorting functionality	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should login 4.Click on sort dropdown 5.Select "Price Low to High"	NA	Products should be sorted correctly	Sorting working correctly	Passed

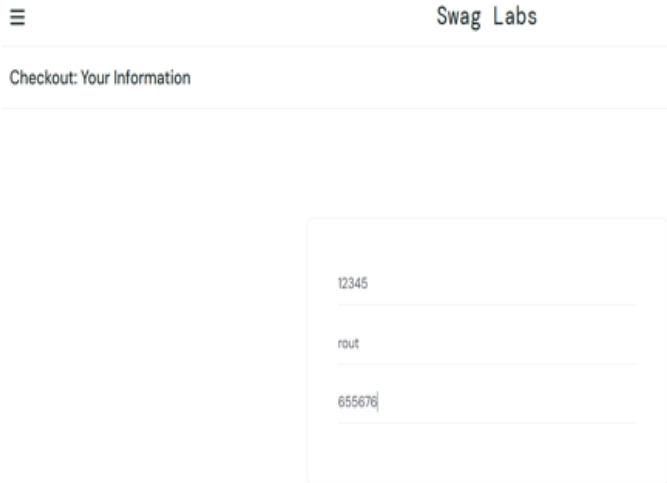
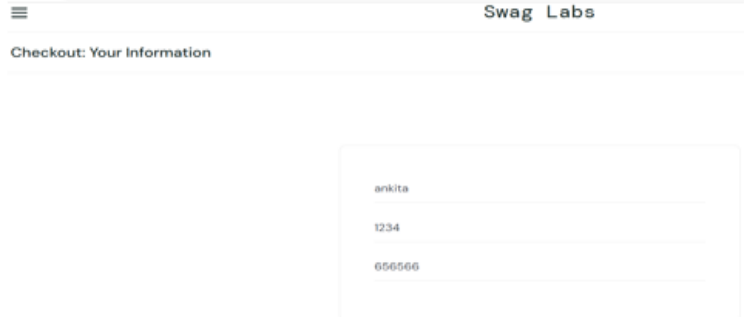
Test Cases

Req_1.5	TC_06	Verify user can proceed to checkout	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3. User should login 4.Add product to cart 5.Click on cart 6.Click on "Checkout" button	NA	User should navigate to checkout information page	Navigation successful	Passed
Req_1.6	TC_07	Verify user enters checkout information	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should be on checkout page 4.Enter First Name 5.Enter Last Name 6.Enter Zip Code 7.Click Continue	Name: Ankita Zip: 751001	User should proceed to overview page	Data accepted successfully	Passed
Req_1.7	TC_08	Verify order placement	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should be on overview page	NA	Order confirmation message should be displayed	Order placed successfully	Passed

Test Cases

Req_1.6	TC_07	Verify user enters checkout information	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should be on checkout page 4.Enter First Name 5.Enter Last Name 6.Enter Zip Code 7.Click Continue	Name: Ankita Zip: 751001	User should proceed to overview page	Data accepted successfully	Passed
Req_1.7	TC_08	Verify order placement	1.User should open Microsoft Edge browser 2. User should navigate to https://www.saucedemo.com/ 3.User should be on overview page 4.Click on Finish button	NA	Order confirmation message should be displayed	Order placed successfully	Passed
Req_1.8	TC_09	Verify logout functionality	1.User should open Microsoft Edge browser 2.User should navigate to https://www.saucedemo.com/ 3. User should login 4.Click on menu icon 5.Click on Logout	NA	User should be logged out and redirected to login page	Logout successful	Passed

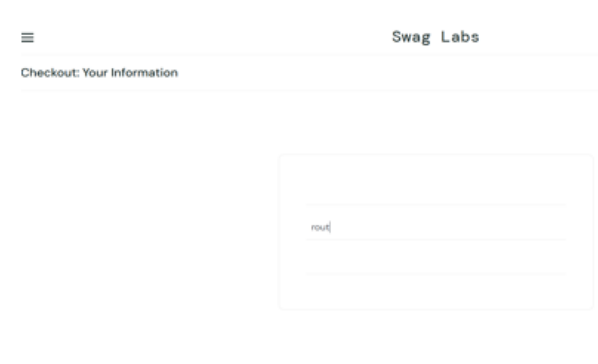
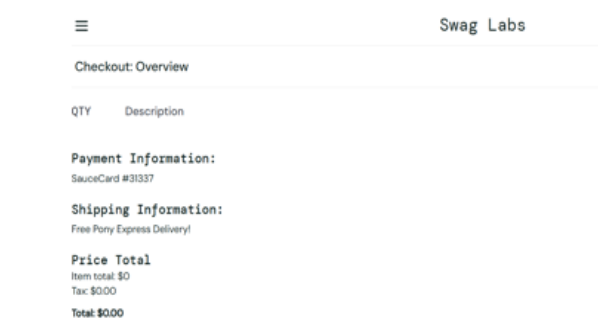
Defect Report

Defect ID	Defect Description	Steps to Reproduce	Priority	Severity	Status	Created By	Assigned To	Screen Shots
1	First Name field accepts numeric values and only Special Characters also	1. Launch browser 2.Open https://www.saucedemo.com/ 3. Login with valid credentials 4. Add product to cart 5. Click Checkout6. Enter numeric value (12345) in First Name field 6. Click Continue 7. Observe System behaviour	High	Critical	New	Ankita Rout	Raghavendra BN	
2	Last Name field accepts Numeric values and Only Special Characters	1. Launch www.google.com 2. Open https://www.saucedemo.com/ 3. Add product to cart 4. Click Checkout 5. Enter 1234 in Last Name 6. Click Continue 7. Observe System behaviour	High	Major	New	Ankita Rout	Raghavendra BN	

Defect Report

3	Zip Code field accepts alphabets	1. Launch www.google.com 2. Open https://www.saucedemo.com/ 3. Add product to cart 4. Click Checkout 5. Enter alphabets (ABC) in Zip Code 6. Click Continue 7. Observe System behaviour	High	Critical	New	Ankita Rout	Raghavendra BN	Swag Labs Checkout: Your Information ankita 1234 ABCD
4	Order can be placed with empty cart	1. Launch www.google.com 2. Open https://www.saucedemo.com/ 3. Login with valid credentials 4. Do not add any product to cart 5. Navigate directly to cart page 6. Click on Checkout button 7. Enter valid user details 8. Click Continue → Finish 9.	High	Critical	New	Ankita Rout	Raghavendra BN	Swag Labs Checkout: Overview QTY Description Payment Information: SauceCard #31337 Shipping Information: Free Pony Express Delivery! Price Total Item total: \$0 Tax: \$0.00 Total: \$0.00 ← Cancel

Defect Report

5	Name and Zip Code fields accept only spaces and still allow order placement	1. Launch browser 2. Open https://www.saucedemo.com/ 3. Login with valid credentials 4. Add product to cart 5. Click Checkout 6. Enter only spaces (" ") in First Name field 7. Enter valid Last Name 8. Enter only spaces (" ") in Zip Code field 9. Click Continue → Finish 10. Observe system behavior	High	Critical	New	Ankita Rout	Raghavendra BN	 
---	---	---	------	----------	-----	-------------	----------------	--

Requirement Traceability Matrix

Req No.	Requirement Description	Test Case ID	Status
R_1.0	The system should allow a user of SwagLabs.com Login Module to login successfully using valid username and password.	TC_01	TC_01 – Passed
R_1.1	The system should not allow login with invalid credentials in SwagLabs Login Module .	TC_02	TC_02 – Passed
R_1.2	The system should redirect the user to the Inventory (Dashboard) Module after successful login.	TC_03	TC_03 – Passed
R_1.3	The system should allow the user of Product Listing Module to view all available products.	TC_04	TC_04 – Passed
R_1.4	The system should allow the user of Product Sorting Module to sort products by price/name.	TC_05	TC_05 – Passed
R_1.5	The system should allow the user of Add to Cart Module to add products to cart successfully.	TC_06	TC_06 – Passed
R_1.6	The system should allow the user of Shopping Cart Module to view selected items.	TC_07	TC_07 – Passed
R_1.7	The system should allow the user of Remove from Cart Module to remove items from cart.	TC_08	TC_08– Passed
R_1.8	The system should allow the user of Checkout Information Module to enter First Name, Last Name, and Zip Code.	TC_09	TC_09– Passed
R_1.9	The system should validate mandatory fields in checkout form before proceeding.	TC_10	TC_10-Failed

<
>
Test Scenarios
Test Case-CaseStudy
Defect Report-CaseStudy
RTM-CaseStudy
+

Requirement Traceability Matrix

[illegible]

Peer Review



Ankita Rout <routankita293@gmail.com>

to Raghavendra ▾

10:14 AM (10 hours ago)

Hi Sir,

I have shared the test deliverables for peer review. Please find attached the Test Plan, Test Scenarios, Test Cases, Defect Report and RTM.

Kindly review and share your feedback or suggestions at your convenience.

Thank you for your time and support.

Best Regards,

Ankita Rout



Raghavendra Bn

to me ▾

10:17 AM (10 hours ago)



Hi Ankita,

Thank you very much for sending across the test deliverables. I have successfully received all the documents, including the Test Plan, Test Scenarios, Test Cases, Defect Report, and the Requirements Traceability Matrix (RTM). I appreciate the effort and attention to detail that has gone into preparing these materials.

I will thoroughly reviewed each of the documents to ensure that all aspects of the testing process have been adequately covered. My review will focus on the completeness and accuracy of the test cases, the clarity of the test scenarios, and the traceability between requirements and test cases as documented in the RTM. Additionally, I will examine the defect report to understand the issues identified so far and how they have been addressed.

If there are any particular sections or areas within these documents that you would like me to pay special attention to, please let me know. Your input will help me provide more targeted and constructive feedback.

I have completed my review, I will compile my observations and share detailed feedback with you. If I have any questions or require further clarification during the review process, I will reach out accordingly.

Thank you again for your hard work and for keeping me in the loop. Looking forward to collaborating further to ensure the quality and success of this project.

Good To Go

Best regards,

Raghavendra BN



SECTION 2

Automation Testing

Running test cases automatically using tools & scripts to verify the application works correctly - built on Automation Exercise.

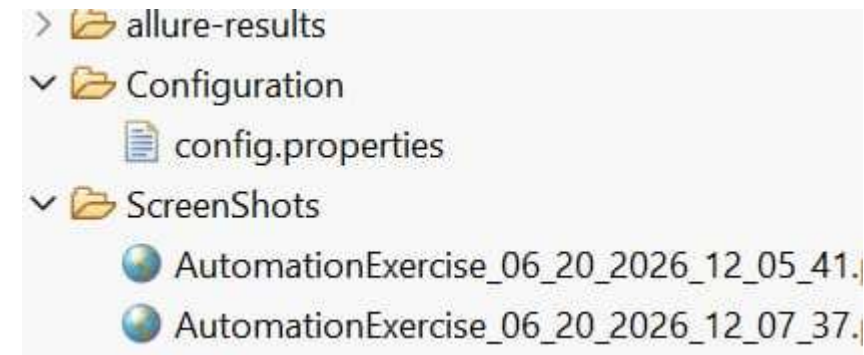
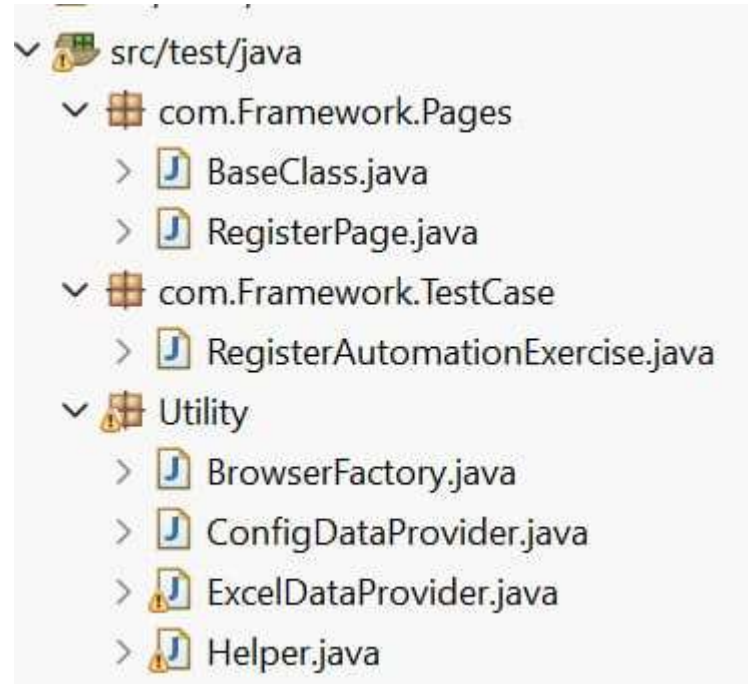
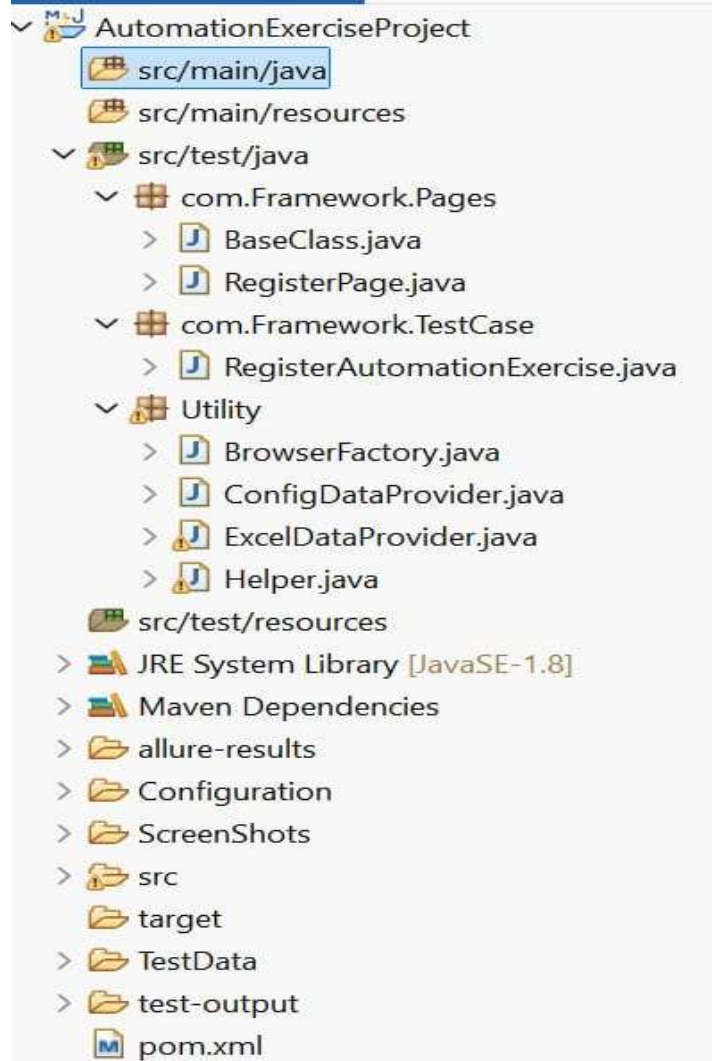
Increases speed, accuracy, and reliability for repetitive and regression testing.

Tools & Setup

JDK(JAVA)	21(Latest Stable)
Selenium	4.39.0
TestNG	7.11
Apache POI	5.5.1
Postman	11.70.0
Eclipse	2022

Allure Reports	2.17.0
Apache JMeter	5.6.3
MySQL Workbench	8.0.34
Jenkins	LTS 2.414.3
JDBC	

Folder Structure



pom.xml

```

<dependencies>
  <!-- Selenium -->
  <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.44.0</version>
    <scope>compile</scope>
  </dependency>

  <!-- TestNG -->
  <dependency>
    <groupId>org.testng</groupId>
    <artifactId>testng</artifactId>
    <version>7.12.0</version>
    <scope>test</scope>
  </dependency>

  <!-- OpenCSV -->
  <dependency>
    <groupId>au.com.bytecode</groupId>
    <artifactId>opencsv</artifactId>
    <version>2.4</version>
    <scope>compile</scope>
  </dependency>

  <!-- Apache POI -->
  <dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi</artifactId>
    <version>5.5.1</version>
    <scope>compile</scope>
  </dependency>

  <!-- Apache POI OOXML -->
  <dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi-ooxml</artifactId>
    <version>5.5.1</version>
    <scope>compile</scope>
  </dependency>

  <!-- Allure Report -->
  <dependency>
    <groupId>io.qameta.allure</groupId>
    <artifactId>allure-testng</artifactId>
    <version>2.35.2</version>
    <scope>compile</scope>
  </dependency>

  <!-- Maven Surefire Report Plugin -->
  <dependency>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-report-plugin</artifactId>
    <version>3.5.6</version>
    <scope>compile</scope>
  </dependency>
</dependencies>

```

Helper.java

```
1 package Utility;
2
3 import java.io.File;
4
5 public class Helper {
6     public static void capturedScreenShot(WebDriver driver)
7     {
8         try {
9             TakesScreenshot ts= (TakesScreenshot) driver;
10             File source = ts.getScreenshotAs(OutputType.FILE);
11             FileUtils.copyFile(source, new File("./ScreenShots/AutomationExercise_"+ getCurrentDateTime()+
12             System.out.println("captured Screenshot - on failure");
13
14         } catch (Exception e) {
15             System.out.println("Exception while taking screenshots "+e.getMessage());
16         }
17     }
18
19     public static String getCurrentDateTime() {
20         DateFormat customFormat = new SimpleDateFormat("MM_dd_yyyy_HH_mm_ss");
21         Date currentdate = new Date();
22         return customFormat.format(currentdate);
23     }
24 }
25 }
```

BrowserFactory.java

```
browserfacto... < ConfigDataF... ExcelDataFIO... Helper.java RegisterAuto... AutomationEX...
1 package Utility;
2
3 import org.openqa.selenium.WebDriver;
4 import org.openqa.selenium.chrome.ChromeDriver;
5 import org.openqa.selenium.edge.EdgeDriver;
6 import org.openqa.selenium.firefox.FirefoxDriver;
7
8 public class BrowserFactory {
9     static WebDriver driver;
10    public static WebDriver BrowserOptions(WebDriver driver, String browserName,String AppUrl) {
11        if(browserName.equalsIgnoreCase("chrome")) {
12            driver = new ChromeDriver();
13        }else if(browserName.equalsIgnoreCase("firefox")) {
14            driver = new FirefoxDriver();
15        }else if(browserName.equalsIgnoreCase("edge")) {
16            driver = new EdgeDriver();
17        } else {
18            System.out.println("Browser does not support....");
19        }
20        driver.manage().window().maximize();
21        driver.get(AppUrl);
22        return driver;
23    }
24
25    public static void quitBrowser(WebDriver driver) {
26        driver.quit();
27    }
28
29 }
30 }
```

ConfigDataProvider.java

```
package Utility;

import java.io.File;

public class ConfigDataProvider {
    Properties pro;
    public ConfigDataProvider() {
        File src = new File("./Configuration/config.properties");
        try {
            FileInputStream fis = new FileInputStream(src);
            pro = new Properties();
            pro.load(fis);
        } catch (Exception e) {
            System.out.println("Unable to load excel file");
        }
    }
    public String getBrowser() {
        return pro.getProperty("Browser");
    }
    public String getAppURL() {
        return pro.getProperty("AppUrl");
    }
}
```

ExcelDataProvider.java

```
10 public class ExcelDataProvider {
11
12     XSSFWorkbook wb;
13
14     public ExcelDataProvider() {
15         try {
16             String path = System.getProperty("user.dir") + "/TestData/DataTest.xlsx";
17
18             FileInputStream fis = new FileInputStream(path);
19             wb = new XSSFWorkbook(fis);
20
21         } catch (Exception e) {
22             System.out.println("Unable to load Excel file " + e.getMessage());
23         }
24     }
25
26     public String getCellData(String sheetName, int row, int column) {
27
28         Cell cell = wb.getSheet(sheetName).getRow(row).getCell(column);
29
30         if (cell.getCellType() == CellType.STRING) {
31             return cell.getStringCellValue();
32
33         } else if (cell.getCellType() == CellType.NUMERIC) {
34             return String.valueOf((int) cell.getNumericCellValue());
35
36         } else if (cell.getCellType() == CellType.BOOLEAN) {
37             return String.valueOf(cell.getBooleanCellValue());
38         }
39
40         return "";
```

BaseClass.java

```
package com.Framework.Pages;

import org.openqa.selenium.WebDriver;

public class BaseClass {
    public WebDriver driver;
    public Utility.ExcelDataProvider excel;
    public ConfigDataProvider config;

    @BeforeSuite
    public void SetUp() {
        excel = new ExcelDataProvider();
        config = new ConfigDataProvider();
    }

    @BeforeClass
    public void BrowserTest() {
        driver = BrowserFactory.BrowserOptions(driver, config.getBrowser(), config.getAppURL());
    }

    @AfterClass
    public void tearDown() {
        BrowserFactory.quitBrowser(driver);
    }

    @AfterMethod
    public void tearDownMethod(ITestResult result) {
        if(result.getStatus() == ITestResult.FAILURE) {
            Helper.capturedScreenShot(driver);
        }
    }
}
```

3.3 CODE FILES

RegisterPage.java

```
package com.Framework.Pages;

import org.openqa.selenium.WebDriver;

public class RegisterPage {

    WebDriver driver;

    // Create Constructor
    public RegisterPage(WebDriver driver) {
        this.driver = driver;
        PageFactory.initElements(driver, this);
    }

    // ===== Signup Page =====

    @FindBy(name = "name")
    WebElement signupName;

    @FindBy(xpath = "//input[@data-qa='signup-email']")
    WebElement signupEmail;

    @FindBy(xpath = "//button[@data-qa='signup-button']")
    WebElement signupBtn;

    // ===== Account Info =====

    @FindBy(id = "id_gender1")
    WebElement mrRadio;
```

```
36     @FindBy(id = "id_gender2")
37     WebElement mrsRadio;
38
39     @FindBy(id = "password")
40     WebElement password;
41
42     // DOB dropdowns (we will use Select manually)
43
44     // ===== Address Info =====
45
46     @FindBy(id = "first_name")
47     WebElement firstName;
48
49     @FindBy(id = "last_name")
50     WebElement lastName;
51
52     @FindBy(id = "company")
53     WebElement company;
54
55     @FindBy(id = "address1")
56     WebElement address1;
57
58     @FindBy(id = "address2")
59     WebElement address2;
60
61     @FindBy(id = "state")
62     WebElement state;
63
64     @FindBy(id = "city")
65     WebElement city;
66
```

RegisterPage.java

```

67 @FindBy(id = "zipcode")
68 WebElement zipcode;
69
70 @FindBy(id = "mobile_number")
71 WebElement mobile;
72
73 @FindBy(xpath = "//button[@data-qa='create-account']")
74 WebElement createAccountBtn;
75
76 // ===== METHODS =====
77
78 // Signup Step
79 public void enterSignupDetails(String name, String email) {
80     signupName.sendKeys(name);
81     signupEmail.sendKeys(email);
82     signupBtn.click();
83 }
84
85 // Select Title
86 public void selectTitle(String title) {
87     if (title.equalsIgnoreCase("Mr")) {
88         mrRadio.click();
89     } else {
90         mrsRadio.click();
91     }
92 }
93
94 // Enter Password
95 public void enterPassword(String pass) {
96     password.sendKeys(pass);
97 }

```

```

CorningData... ExcelDataTo... Helper.java RegisterAuto... BaseClass.java
10 public void selectDOB(String day, String month, String year) {
11
12     Select dayDrop = new Select(driver.findElement(By.id("days")));
13     dayDrop.selectByVisibleText(day);
14
15     Select monthDrop = new Select(driver.findElement(By.id("months")));
16     monthDrop.selectByVisibleText(month);
17
18     Select yearDrop = new Select(driver.findElement(By.id("years")));
19     yearDrop.selectByVisibleText(year);
20 }
21
22 // Address Details
23 public void enterAddressDetails(String fName, String lName, String comp,
24     String addr1, String addr2,
25     String stateVal, String cityVal,
26     String zip, String mob) {
27
28     firstName.sendKeys(fName);
29     lastName.sendKeys(lName);
30     company.sendKeys(comp);
31     address1.sendKeys(addr1);
32     address2.sendKeys(addr2);
33     state.sendKeys(stateVal);
34     city.sendKeys(cityVal);
35     zipcode.sendKeys(zip);
36     mobile.sendKeys(mob);
37 }
38
39 // Click Create Account
40 public void clickCreateAccount() {
41     createAccountBtn.click();
42 }

```

RegisterAutomationExercise.java

```

1  package com.Framework.TestCase;
2
3  import org.openqa.selenium.support.PageFactory;
4  import org.testng.annotations.Test;
5
6  import com.Framework.Pages.BaseClass;
7  import com.Framework.Pages.RegisterPage;
8
9  import Utility.ExcelDataProvider;
10
11  Run ALL
12  public class RegisterAutomationExercise extends BaseClass {
13
14      @Test
15      Run / Debug
16      public void registerApp() throws Exception {
17
18          ExcelDataProvider excel = new ExcelDataProvider();
19          RegisterPage page = PageFactory.initElements(driver, RegisterPage.class);
20
21          // ===== SIGNUP =====
22          page.enterSignupDetails(
23              excel.getCellData("Sheet1", 0, 1),    // Name
24              excel.getCellData("Sheet1", 0, 2)    // Email
25          );
26
27          Thread.sleep(3000);
28
29          // ===== ACCOUNT =====
30          page.selectTitle(
31              excel.getCellData("Sheet1", 0, 0)    // Title
32          );

```

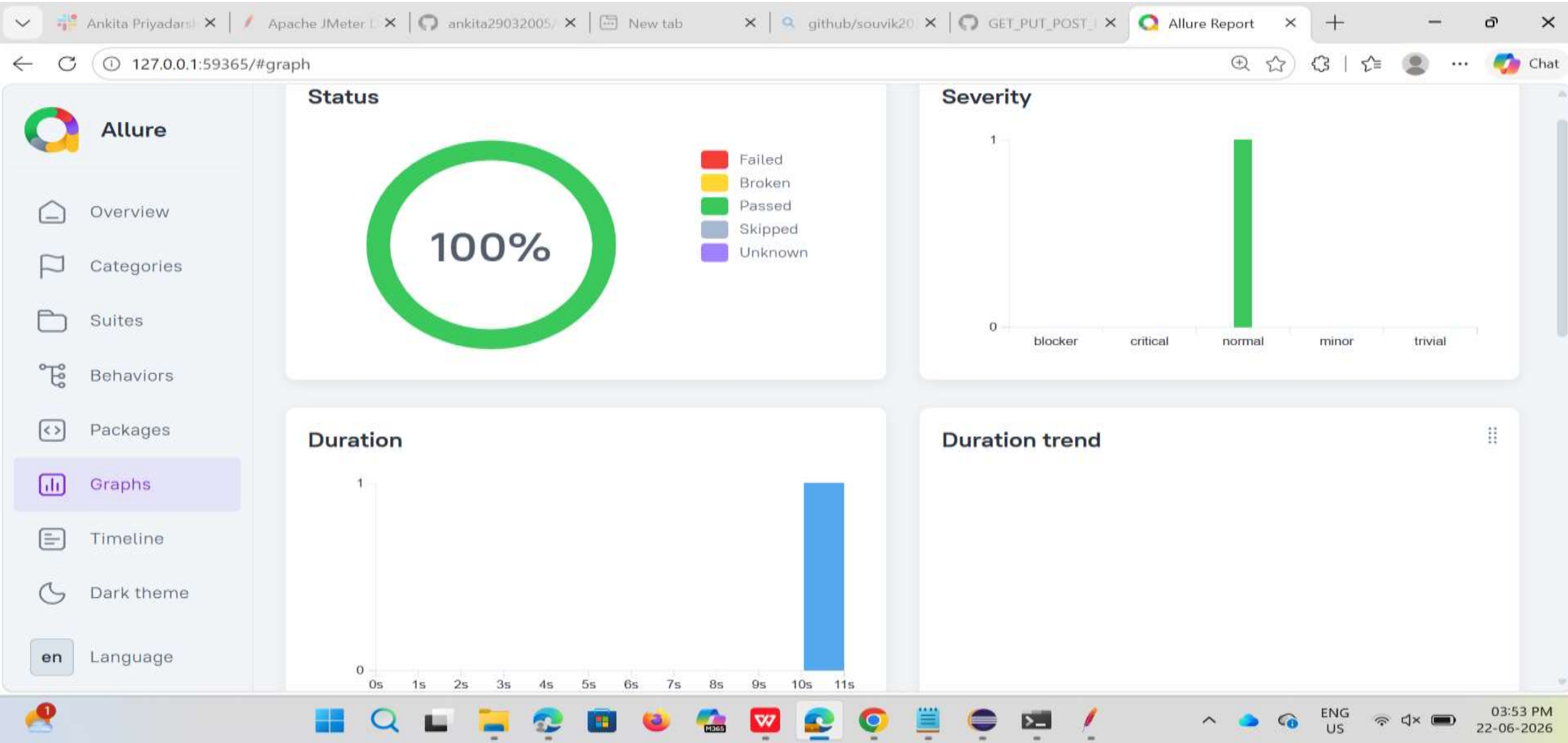
RegisterAutomationExercise.java

```

29         excel.getCellData("Sheet1", 0, 0) // Title
30     );
31
32     page.enterPassword(
33         excel.getCellData("Sheet1", 0, 3) // Password
34     );
35
36     page.selectDOB(
37         excel.getCellData("Sheet1", 0, 4), // Day
38         excel.getCellData("Sheet1", 0, 5), // Month
39         excel.getCellData("Sheet1", 0, 6) // Year
40     );
41
42     // ===== ADDRESS =====
43     page.enterAddressDetails(
44         excel.getCellData("Sheet1", 0, 9), // FirstName
45         excel.getCellData("Sheet1", 0, 10), // LastName
46         excel.getCellData("Sheet1", 0, 11), // Company
47         excel.getCellData("Sheet1", 0, 12), // Address1
48         excel.getCellData("Sheet1", 0, 13), // Address2
49         excel.getCellData("Sheet1", 0, 15), // State
50         excel.getCellData("Sheet1", 0, 16), // City
51         excel.getCellData("Sheet1", 0, 17), // Zip
52         excel.getCellData("Sheet1", 0, 18) // Mobile
53     );
54
55     // ===== FINAL =====
56     page.clickCreateAccount();
57
58     Thread.sleep(3000);
59 }
60 }

```

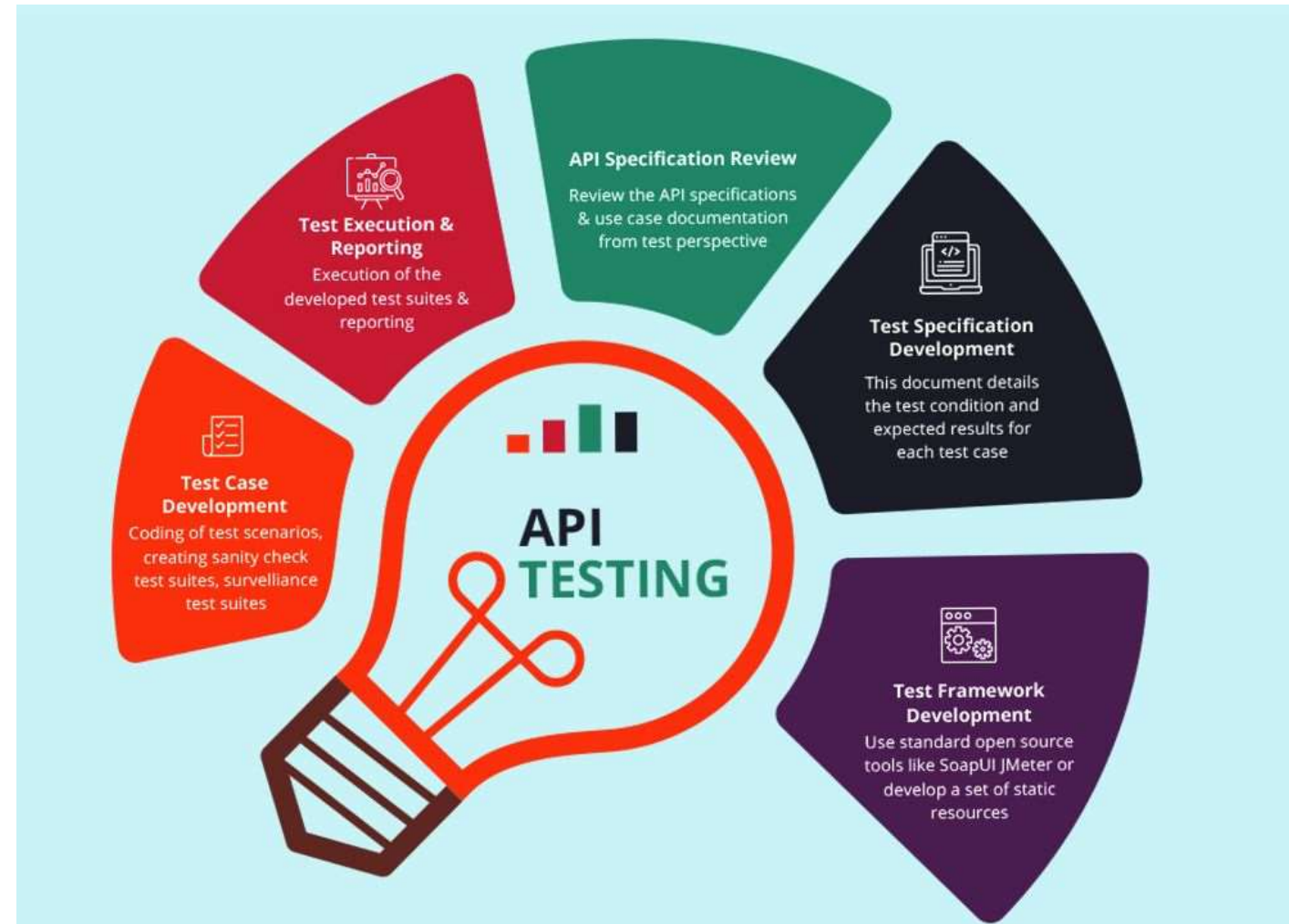
Allure Report



API Testing

API testing is the process of testing application programming interfaces to ensure they function correctly, securely, and efficiently.

It focuses on validating the communication between different software components by checking requests, responses, status codes, and data accuracy without relying on the user interface.



4.1 POSTMAN

Postman

 Mini_Project > GETRequest

GET

{{baseUrl}}

Send

Docs

Params

Authorization

Headers 8

Body

Scripts

Tests

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

200 OK · 866 ms · 51.69 KB · Save Response

HTML

Preview

Visualize

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   <meta name="description" content="This is for automation practice">
8   <meta name="robots" content="index" />
9   <meta name="keywords"
10     content="Automation Exercise / Automation Practice / Practice website for
```

Cloud Ag

 Mini_Project > POST Request

POST

→ {{baseUrl}} login

Send

Docs

Params

Authorization

Headers 9

Body

Scripts

Tests

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

200 OK · 771 ms · 394 B · Save Response

JSON

Preview

Visualize

```
1 {
2   "status": "success",
3   "data": {
4     "id": 6296
5   },
6   "message": "Successfully! Record has been added."
7 }
```

4.2 JMETER REPORT

JMeter

AutomationExercise.jmx (C:\AutomationExercise_jmeter\AutomationExercise.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

AutomationExercise

TestCases-HTTP Request

Constant Timer

Login-HTTP Request

Constant Timer

Cart-HTTP Request

Constant Timer

product-HTTP Request

Constant Timer

Home-HTTP Request

Constant Timer

Contact-HTTP Request

Constant Timer

ApiTesting-HTTP Request

Constant Timer

BrandPolo-HTTP Request

Constant Timer

BrandH&M-HTTP Request

Viewproducts-HTTP Request

Constant Timer

Random Controller

View Results Tree

Aggregate Graph

HTTP Request

Name: TestCases-HTTP Request

Comments:

Basic Advanced

Web Server

Protocol [http]: https Server Name or IP: automationexercise.com Port Number:

HTTP Request

GET Path: /test_cases Content encoding:

Redirect Automatically

Follow Redirects

Use KeepAlive

Use multipart/form-data

Browser-compatible headers

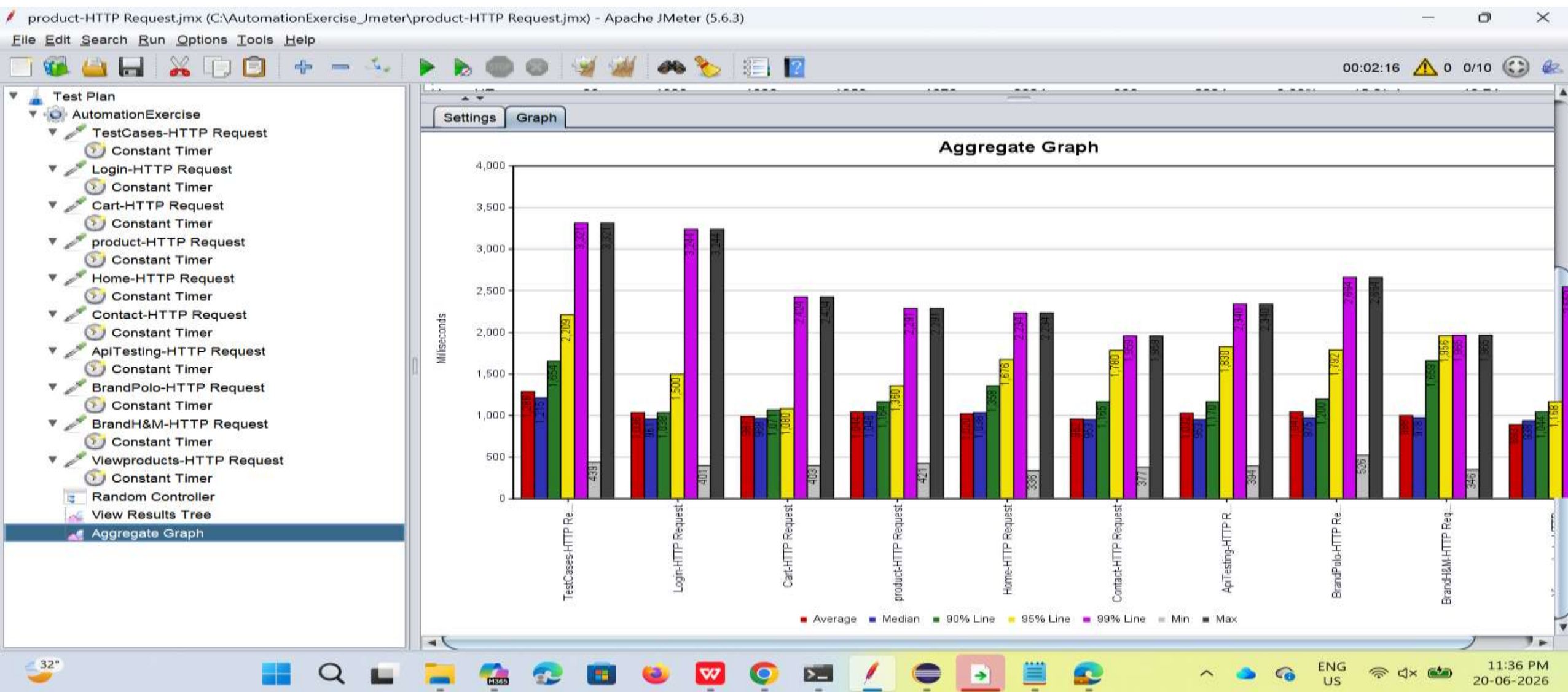
Parameters Body Data Files Upload

Send Parameters With the Request:

Name:	Value	URL Encode?	Content-Type	Include Equals?
-------	-------	-------------	--------------	-----------------

Detail Add Add from Clipboard Delete Up Down

Aggregate Graph



HTML Report

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\ApacheJmeterInstallation\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>jmeter -n -t "C:\AutomationExercise_Jmeter\AutomationExercise.jmx" -l "C:\AutomationExercise_Jmeter\NONGUI\AutomationExercise.csv" -e -o "C:\AutomationExercise_Jmeter\html\Reports"
```

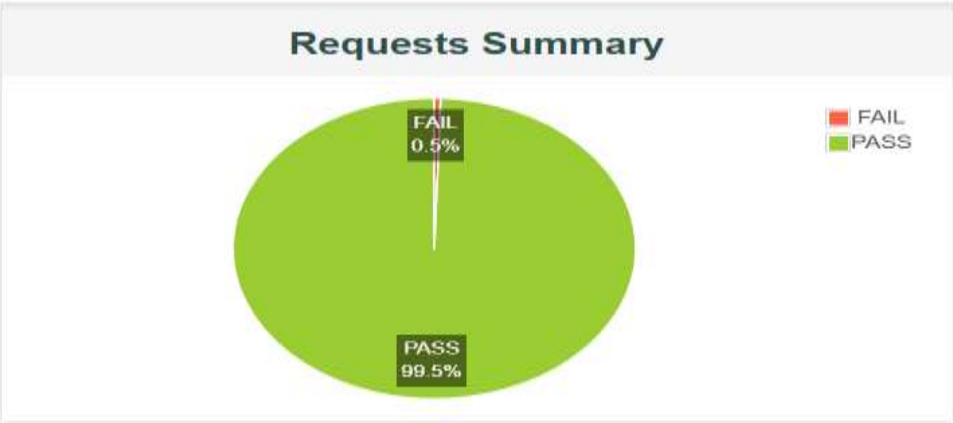
Dashboard

Charts

Customs Graphs

Test and Report information	
Source file	"AutomationExercise.csv"
Start Time	"20/06/26, 11:19 pm"
End Time	"20/06/26, 11:22 pm"
Filter for display	""

APDEX (Application Performance Index)			
Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.495	500 ms	1 sec 500 ms	Total
0.275	500 ms	1 sec 500 ms	TestCases-HTTP Request
0.400	500 ms	1 sec 500 ms	Home-HTTP Request
0.475	500 ms	1 sec 500 ms	product-HTTP Request



JMeter Report

Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	200	1	0.50%	1121.62	356	4892	1021.50	1890.90	2367.25	3795.97	1.61	37.55	0.21
ApiTesting-HTTP Request	20	0	0.00%	962.40	410	2045	970.00	1459.10	2015.80	2045.00	0.30	6.08	0.04
BrandH&M-HTTP Request	20	0	0.00%	1091.20	411	3124	998.00	1780.40	3057.35	3124.00	0.29	4.78	0.04
BrandPolo-HTTP Request	20	0	0.00%	1045.30	392	1902	1139.50	1652.90	1890.80	1902.00	0.30	5.27	0.04
Cart-HTTP Request	20	0	0.00%	898.80	421	1221	1007.50	1196.30	1220.15	1221.00	0.30	2.47	0.04
Contact-HTTP Request	20	0	0.00%	943.55	375	2107	958.50	1595.10	2081.75	2107.00	0.30	2.96	0.04
Home-HTTP Request	20	1	5.00%	1303.70	439	3423	1044.50	2339.50	3369.50	3423.00	0.30	14.90	0.04
Login-HTTP Request	20	0	0.00%	916.40	356	1665	959.00	1500.00	1657.70	1665.00	0.30	2.30	0.04

4.3. Rest Assured

Rest Assured

Get Request

```
1 package com.restAssured.Demo;
2+ import static io.restassured.RestAssured.*;
7
8 public class BDDGetRequest3 {
9
10-     public static void main(String[] args) {
11         Response response = given().when().get("https://dummy.restapiexample.com/api/v1/employee/1").then().extract().r
12         System.out.println("Status code "+response.getStatusCode());
13
14         System.out.println("complete response: ");
15         System.out.println(response.asPrettyString());
16
17         System.out.println("Response body: ");
18         System.out.println(response.getBody().asPrettyString());
19
20         System.out.println("Status line: "+ response.getStatusLine());
21
22         System.out.println("Content Type "+response.getContentType());
23
24         System.out.println("Response Time "+ response.getTime()+ "ms");
25
26     }
```

Problems @ Javadoc Declaration Console × Results of running method CompletePutPostTest.testPostRequestComplete

terminated> BDDGetRequest3 [Java Application] C:\Users\Ankita.Rout\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\

Status code 200

complete response:

```
{
  "status": "success",
  "data": {
    "id": 1,
    "employee_name": "Tiger Nixon",
    "employee_salary": "320800".
```

Rest Assured

Post Request

```

1 package com.example.tests;
2+ import io.restassured.RestAssured;
10
11 Run ALL
11 public class CompletePutPostTest {
12     private static final String BASE_URL = "https://jsonplaceholder.typicode.com";
13     private JSONObject testUser;
14     private int createdUserId;
15
16     @BeforeClass
17- public void setup() {
18         RestAssured.baseURI = BASE_URL;
19         // Common test data
20         testUser = new JSONObject();
21         testUser.put("name", "Test User");
22         testUser.put("email", "raghu.asteapahead@gmail.com");
23         testUser.put("username", "raghavendra");
24     }
25     @Test(priority = 1)
26- Run / Debug
26- public void testPostRequestComplete() {
27         System.out.println("\n\n===== POST REQUEST - TRADITIONAL =====");

```

Problems
 Javadoc
 Declaration
 Console
 Results of running class CompletePutPostTest

```

<terminated> CompletePutPostTest [TestNG] C:\Users\Ankita.Rout\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11
--- POST Response Details ---
1. Status Code: 201
2. Complete Response: {
  "name": "Test User",
  "id": 11,
  "email": "raghu.asteapahead@gmail.com",
  "username": "raghavendra"
}
3. Response Body: {

```

Rest Assured

Put Request

```

Run | Debug
80 public void testPutRequestBDDComplete() {
81     System.out.println("\n\n===== PUT REQUEST - BDD =====");
82     JSONObject updatedPost = new JSONObject();
83     updatedPost.put("id", 1);
84     updatedPost.put("title", "BDD Updated Post");
85     updatedPost.put("body", "Updated using BDD approach");
86     updatedPost.put("userId", 1);
87
88     System.out.println("Request Body: " + updatedPost.toString(2));
89
90     given().header("Content-Type", "application/json")
91         .body(updatedPost.toString())
92         .when().put("/posts/1")
93         .then()
94             .statusCode(200)
95             .statusLine(containsString("200"))
96             .contentType("application/json")
97             .body("title", equalTo("BDD Updated Post"))
98             .body("body", equalTo("Updated using BDD approach"))
99             .body("userId", equalTo(1))
100            .time(LessThan(5000L))

```

Problems @ Javadoc Declaration Console × Results of running class CompletePutPostTest

```

terminated> CompletePutPostTest [TestNG] C:\Users\Ankita.Rout\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.f
===== PUT REQUEST - TRADITIONAL =====
Request Body: {
  "phone": "1-555-555-9999",
  "name": "Updated Test User",
  "id": 1,
  "email": "raghu.astepeahead@example.com",
  "username": "updateduser"
}

```

4.4.Cucumber

Cucumber

The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure for 'cucumber.first.example', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'JRE System Library [jdk-21.0.2]', 'Maven Dependencies', 'Futures', 'LoginTest.feature', 'TestCase2.feature', 'testCase3.feature', 'src', 'target', and 'pom.xml'.
- Editor:** Displays the 'LoginTest.feature' file with the following content:

```
1 Feature: Checking login Functionality
2
3   Scenario: Successful Login with Valid Credentials
4     Given User should open Chrome Browser
5     When User should Enter url in Browser
6     When User should Navigate Home Page
7     And Enter UserName and Password in Edit Box
8     And Click On Login PushButton
9     Then Message displayed Login Successfully
```
- Test Results:** Shows the test execution summary:
 - Finished after 8.388
 - Runs: 6/6
 - Errors: 0
 - Failures: 0A green progress bar indicates successful completion. Below this, a 'Failure Trace' section is visible for the test 'Checking login Functionality [Runner: Cucumbe]'.
- Console:** Displays the execution log:

```
<terminated> LoginTest.feature [Cucumber-JVM] C:\Program Files\openjdk-21.0.2\jdk-21.0.2\bin\javaw.exe (22-Jun-2026, 10:46:03 am - 10:46:14 am elapsed 0:00:11) [pid: 26484]
Jun 22, 2026 10:46:08 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 149, returning the closest version; found: 148; Please update to a Selenium version that supports CDP
Navigated to login page
Login Success URL: https://practicetestautomation.com/logged-in-successfully/

1 scenarios (1 passed)
6 steps (6 passed)
0m 9.822s
```

The Windows taskbar at the bottom shows the system date and time as 10:46 AM on 22-06-2026, along with weather information (31°C, Mostly sunny) and various system icons.

4.4.Cucumber

Cucumber

The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure for 'cucumber.first.example', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'JRE System Library [jdk-21.0.2]', 'Maven Dependencies', 'Futures', 'LoginTest.feature', 'TestCase2.feature', 'testCase3.feature', 'src', 'target', and 'pom.xml'.
- Editor:** Displays the 'TestCase2.feature' file with the following content:

```
1 Feature: Failed Test Case
2
3 Scenario: Login With Invalid UserName
4   Given User should open Chrome Browser
5   When User should Enter url in Browser
6   When User should Navigate to Home Page
7   And Enter Invalid UserName and Valid Password in Edit Box
8   And Click On Login PushButton
9   Then Error Message should be displayed
10  And Verify Error Message Text
```
- Test Results:** Shows the test execution summary:
 - Finished after 7.679
 - Runs: 7/7
 - Errors: 0
 - Failures: 0
- Failure Trace:** Displays the failure details for the 'Failed Test Case [Runner: Cucumber]' which took 7.679s.
- Console:** Shows the execution output:

```
<terminated> TestCase2.feature [Cucumber-JVM] C:\Program Files\openjdk-21.0.2\jdk-21.0.2\bin\javaw.exe (22-Jun-2026, 10:46:33 am – 10:46:43 am elapsed 0:00:10) [pid: 15424]
Jun 22, 2026 10:46:38 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 149, returning the closest version; found: 148; Please update to a Selenium version that supports CDP
Error: Your username is invalid!

1 scenarios (1 passed)
7 steps (7 passed)
0m 9.102s
```

The Windows taskbar at the bottom shows the system date and time as 10:46 AM on 22-06-2026, along with weather (31°C Mostly sunny), search, and various application icons.

Git Repositories

Module Name :	GitHub Repositories	Submitted By :	Ankita Rout
No.of Repositories :		Submitted To :	Raghavendra BN
Emp ID:	85009812		

S.No	Date	Project Name	Repository Link
1	03-06-2026	VariableIn_Java,, ScannerClass, PrimitiveData_types, Operators and Assignment, Java_ControlStatements	https://github.com/ankita29032005/BootCamp_03-06-26_java_Testing
2	04-06-2026	Java_AccessModifiers, Constructor_In_Java , Class_And_Object	https://github.com/ankita29032005/BootCamp_04_06_26_Java_Testing
3	05-06-2026	Java_Testing-Inheritance, Abstract, ExceptionHandling Interface	https://github.com/ankita29032005/BootCamp_05_06_26_Java_Testing
4	08-06-2026	SeleniumWebDriver-BrowserTesting	https://github.com/ankita29032005/JavaEclipseProgram-LaunchBrowserTestCase
5	08-06-2026	TestNG-BrowserTesting, SeleniumWebDriver-CaseStudy1-BrowserTesting	https://github.com/ankita29032005/SeleniumWebDriverTestCases
6	09-06-2026	SeleniumWebDriver -CaseStudy -Commands,SeleniumWebDriver -CaseStudy_Locaters-EasyCalculation	https://github.com/ankita29032005/SeleniumWebDriverTestCases
7	10-06-2026	SeleniumWebBrowser _Case Study-DropDownList,Radio Button,ScrollUpDown, capturing the links and image	https://github.com/ankita29032005/SeleniumWebDriverTestCases
8	11-06-2026	SeleniumWebDriver _CaseStudy-POM-Practicetestautomation.,	https://github.com/ankita29032005/MavenProject-PageObjectModuleTestCases
9	11-06-2026	SeleniumWebDriver _CaseStudy-POM-Profilebrightest website	https://github.com/ankita29032005/MavenProject-PageObjectModuleTestCases
10	11-06-2026	SeleniumWebDriver _CaseStudy-POM- chase Website,	https://github.com/ankita29032005/MavenProject-PageObjectModuleTestCases
11	14-06-2026	Assignment1-Selenium Web Driver	https://github.com/ankita29032005/Assignment1-SeleniumWebDriver
12	14-06-2026	Assignment2-Selenium Web Driver	https://github.com/ankita29032005/Assignment2-EasyCalculation
13	14-06-2026	Assignment6-Selenium Web Driver	https://github.com/ankita29032005/Assignment6
14	12-06-2026	Data Driven CSV Files	https://github.com/ankita29032005/DataDrivenCSVFiles
15	12-06-2026	ApachePOI TestCases	https://github.com/ankita29032005/ApachePoiTestCasesAndHybridDrivenFramework
16	12-06-2026	Hybrid driven FrameWork	https://github.com/ankita29032005/ApachePoiTestCasesAndHybridDrivenFramework
17	14-06-2026	Assignment7-Selenium Web Driver	https://github.com/ankita29032005/Assignment7
18	14-06-2026	Assignment8-Selenium Web Driver	https://github.com/ankita29032005/Assignment8
19	14-06-2026	Assignment9-Selenium Web Driver	https://github.com/ankita29032005/Assignment9
20	13-06-2026	Assertion In TestNG	https://github.com/ankita29032005/Assertion-In-TestNG
21	13-06-2026	Parallel TestNG	https://github.com/ankita29032005/AllMavenProjects
22	13-06-2026	Data Provider In TestNG	https://github.com/ankita29032005/Data-Provider-TestNG
23	14-06-2026	Assignment10-Selenium Web Driver	https://github.com/ankita29032005/Assignment10
24	14-06-2026	Assignment11-Selenium Web Driver	https://github.com/ankita29032005/Assignment11
25	14-06-2026	Assignment12-Selenium Web Driver	https://github.com/ankita29032005/Assignment12
26	14-06-2026	Assignment13-Selenium Web Driver	https://github.com/ankita29032005/Assignment13
27	14-06-2026	Assignment14-Selenium Web Driver	https://github.com/ankita29032005/Assignment14
28	17-06-2026	CaseStudy_ Cucumber-Example	https://github.com/ankita29032005/CucumberTestCase-PracticeAutomation
29	15-06-2026	Synchronization - Selenium	https://github.com/ankita29032005/Synchronization-Selenium
30	17-06-2026	JMeter Assignment	https://github.com/ankita29032005/JMeter-Assignment
31	19-06-2026	Rest Assured Test Cases	https://github.com/ankita29032005/Rest-assured-Case-Study

Thank You!

Questions & Discussion

Manual Testing - SwagLabs | Automation Testing - Automation Exercise